

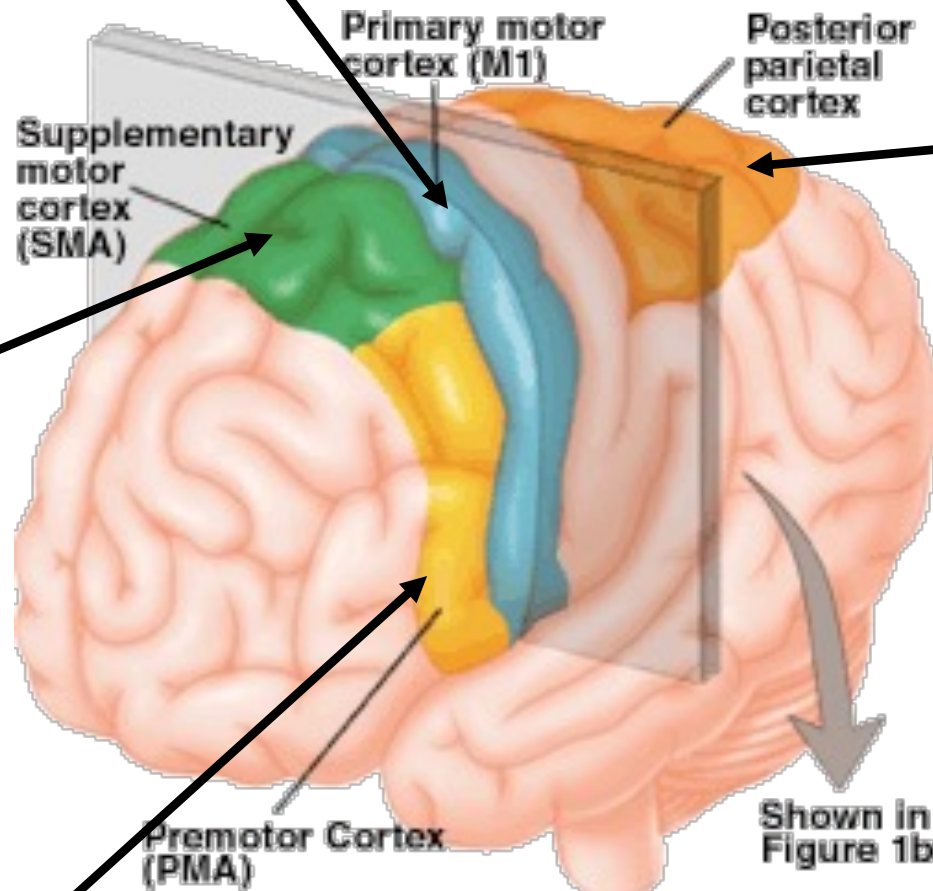
L8

Motor Cognition

Execution

~ 50 ms

**Spatial
Information**



Planning

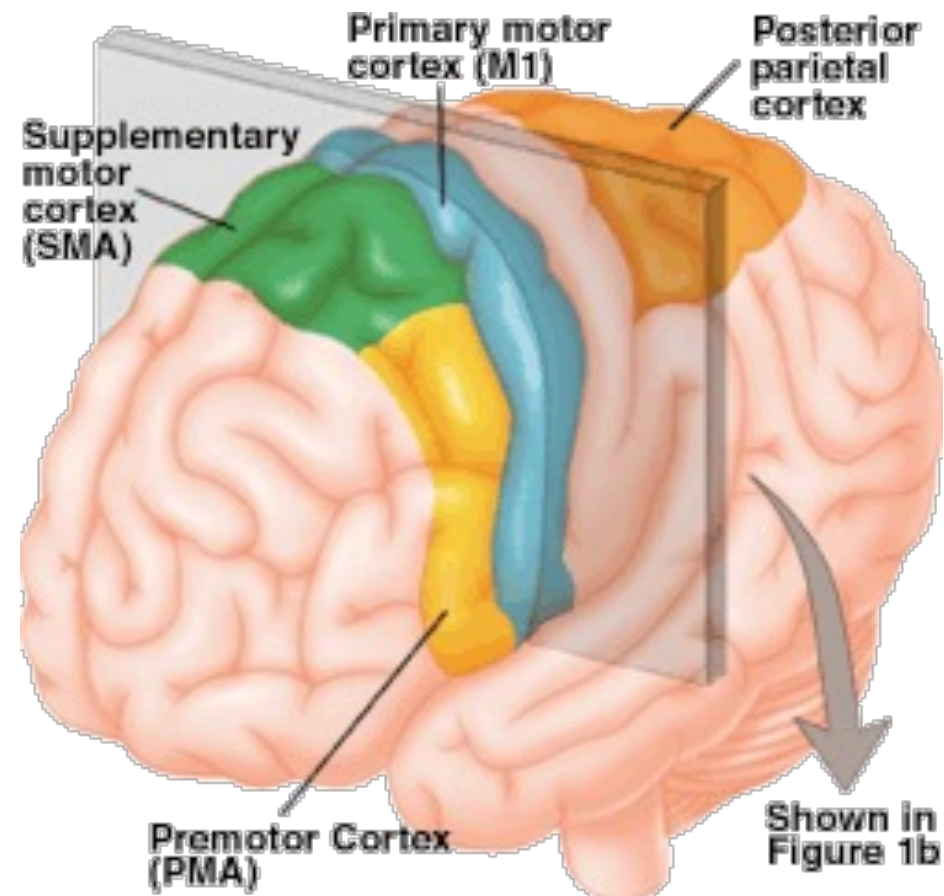
~1 sec

Preparation

~120-150 ms

<https://brainconnection.brainhq.com/2013/03/05/the-anatomy-of-movement/>

Parietal Cortex



<https://brainconnection.brainhq.com/2013/03/05/the-anatomy-of-movement/>

Apraxia

How to prepare the perfect cup of tea?

- **Ideational**
- **Ideomotor**

Spatially relevant behavioral intentions

Hemi-spatial Neglect

Hemineglect, also known as unilateral neglect, hemispatial neglect or spatial neglect, is a common and disabling condition following brain damage in which patients fail to be aware of items to one side of space.

Copying:



Spontaneous drawing:

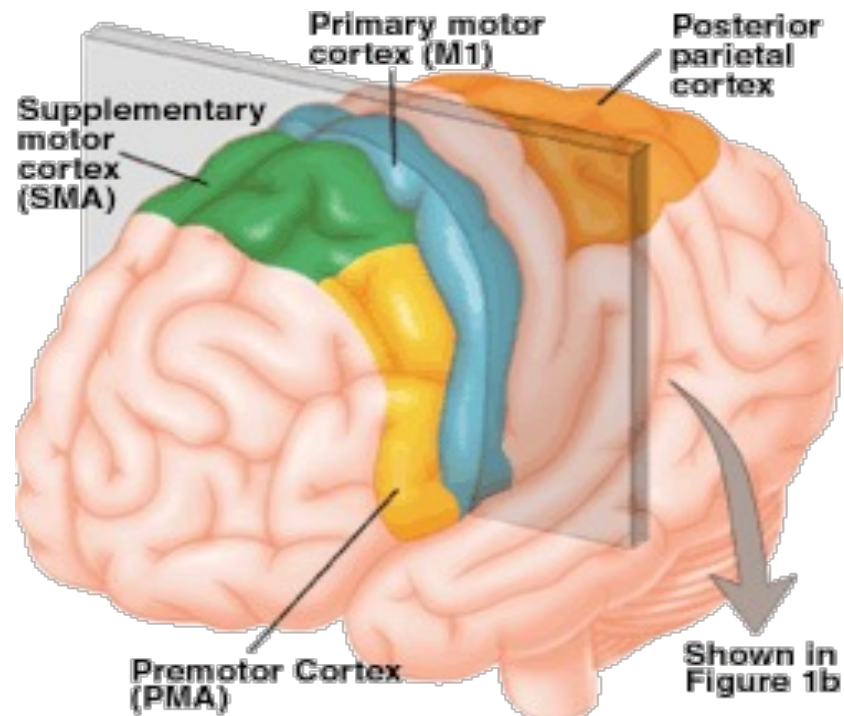


hemianopia: Blindness in one visual field

<https://tactustherapy.com/what-is-left-neglect/>

Occurs in the visual field opposite (Contralateral) to side of brain which is damaged.

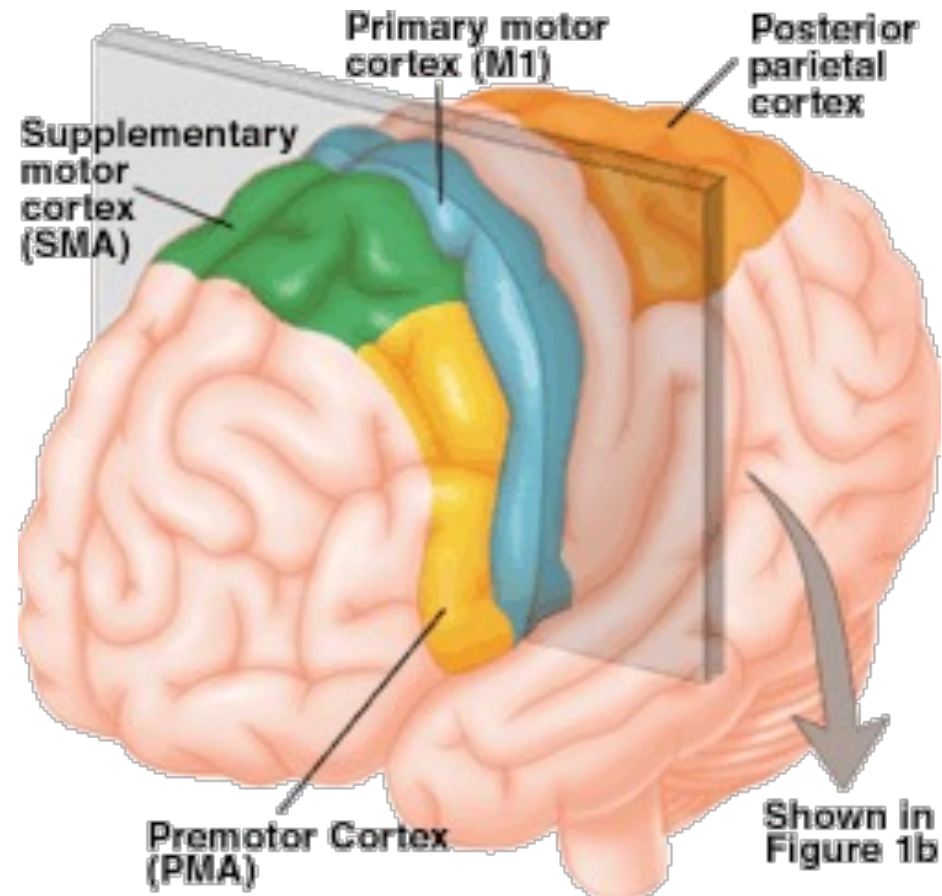
Parietal Cortex



<https://brainconnection.brainhq.com/2013/03/05/the-anatomy-of-movement/>

- Spatially relevant behavioral intentions
- Cross modal integration

Motor Cognition



<https://brainconnection.brainhq.com/2013/03/05/the-anatomy-of-movement/>

Walking



<https://newsroom.uhc.com/health/Move-More-Month.html>

Walking **Gait**

Walking



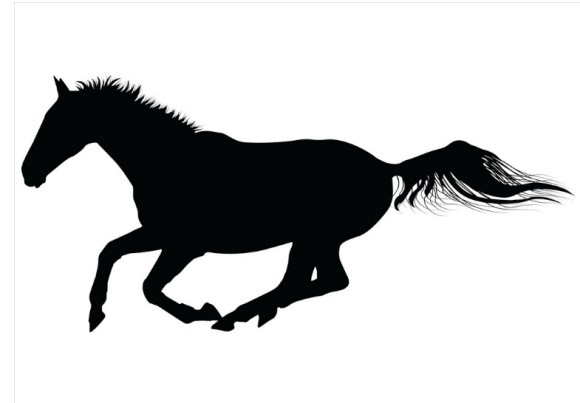
<https://newsroom.uhc.com/health/Move-More-Month.html>

Limping



<https://www.dreamstime.com/photos-images/limping.html>

Galloping



<https://www.vectorstock.com/royalty-free-vector/galloping-horse-silhouette-vector-776365>

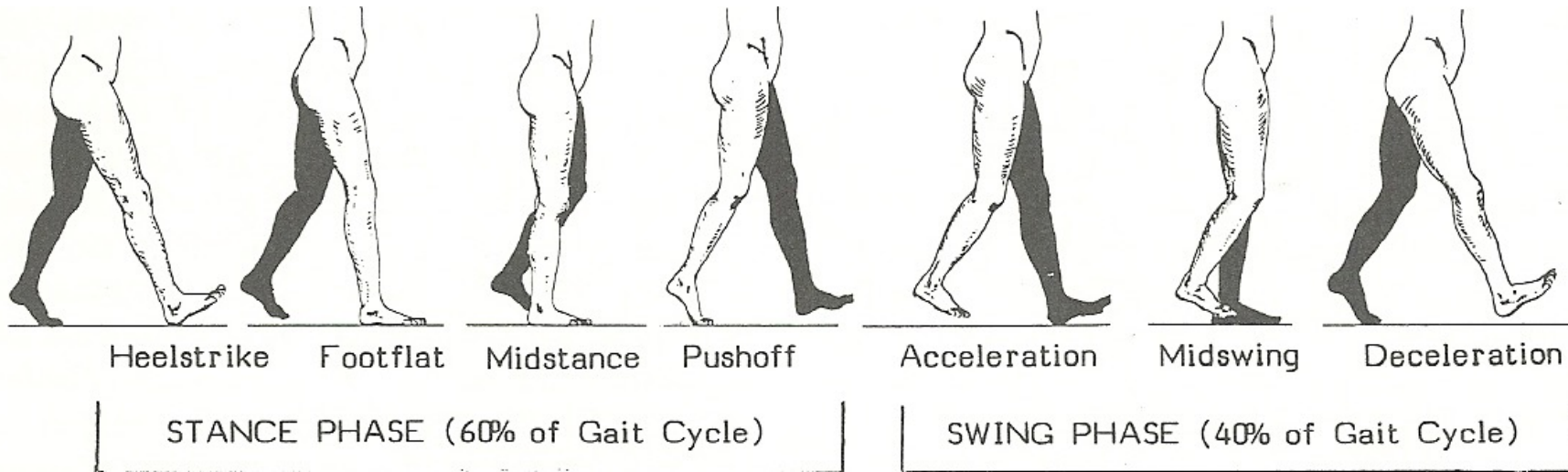
Trotting



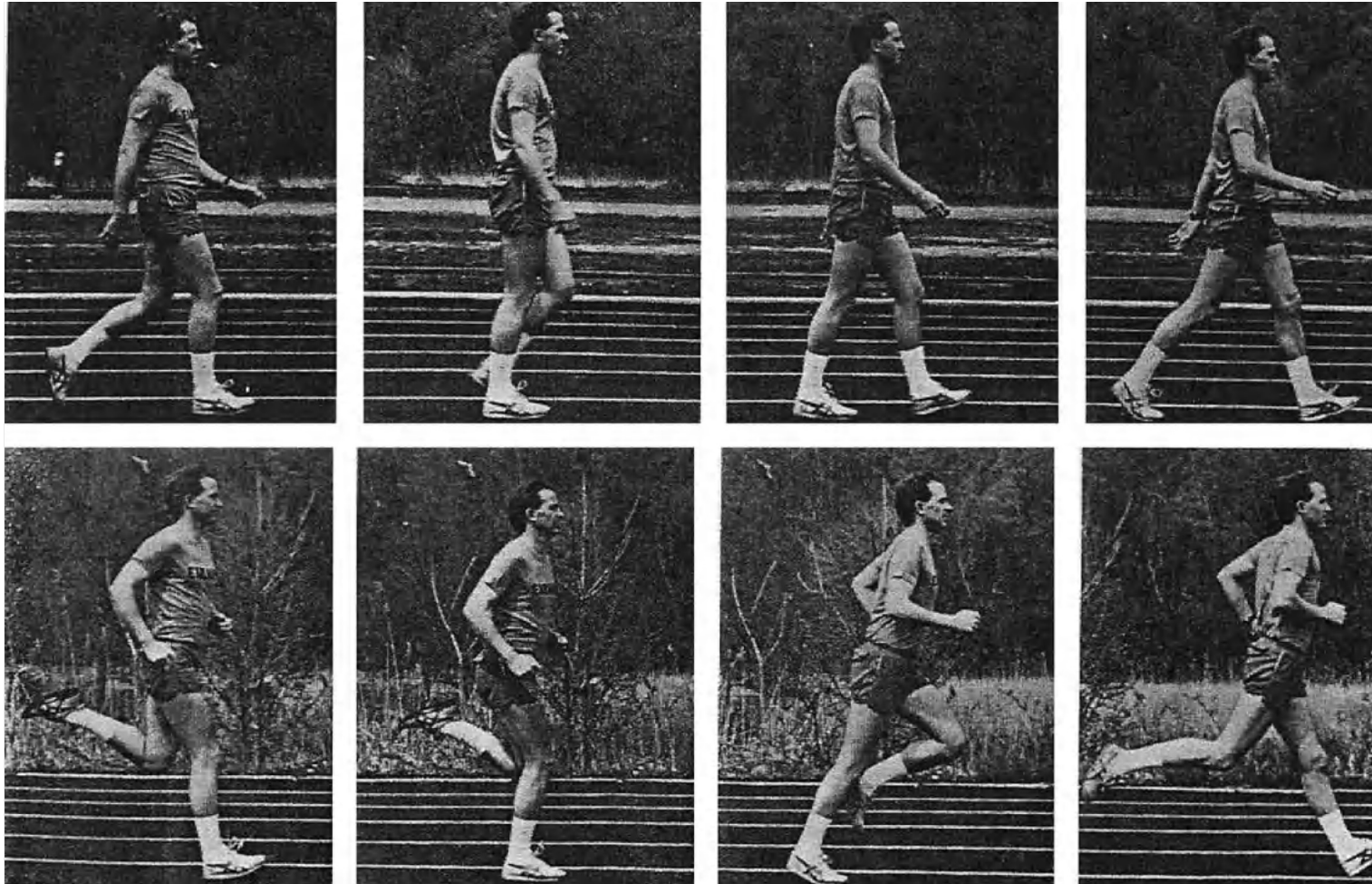
<https://en.wikipedia.org/wiki/Trot>

Way of walking

Walking: Phases

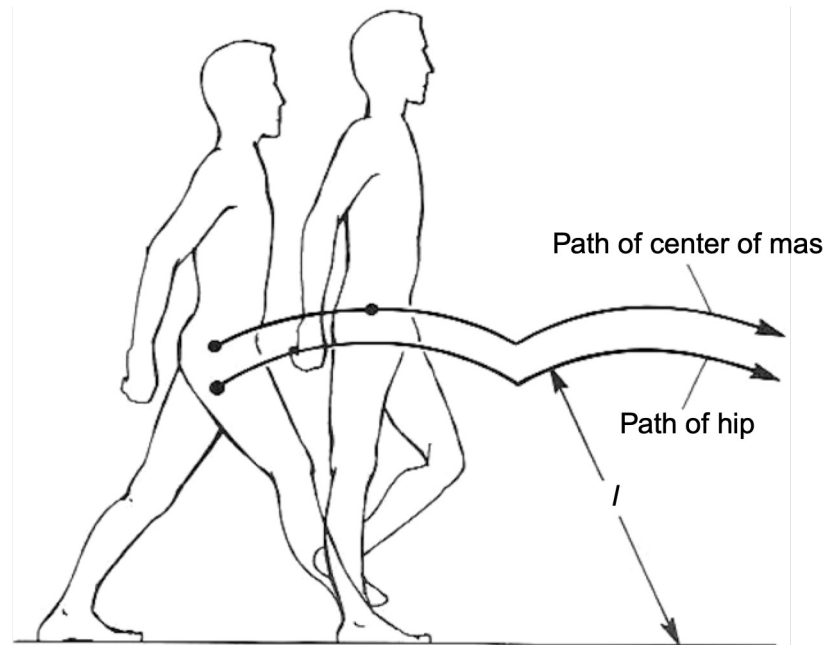


Gait patterns



Gait pattern $\longleftrightarrow$ **Speed of locomotion**

Walking to running



$$m \cdot g \geq m \cdot \left(\frac{v^2}{l} \right)$$

$$g \geq \frac{v^2}{l}$$

$$v \leq \sqrt{g \cdot l}$$

Physical constraints to the speed of locomotion and stance

Walking to running

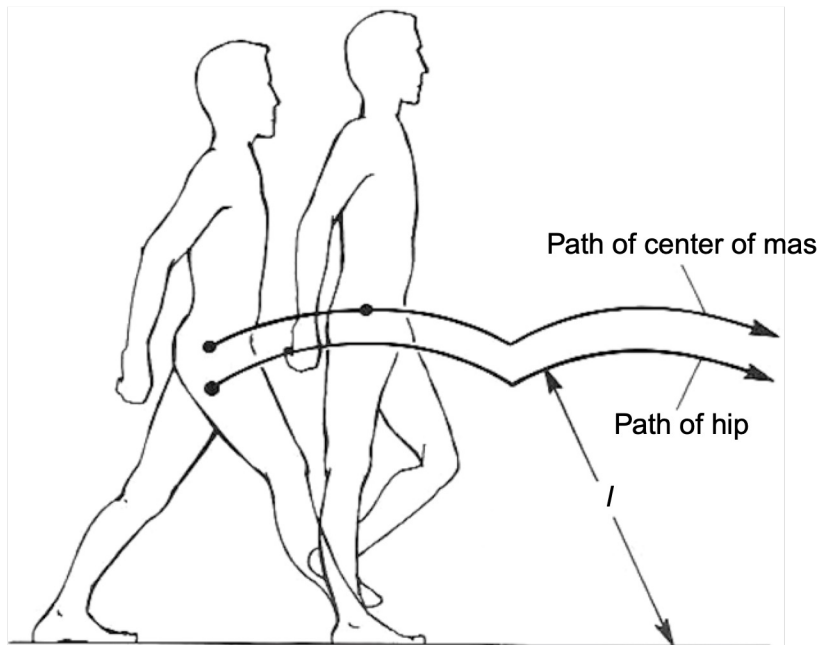


<https://glycat.com/angryringedalligatorgar>

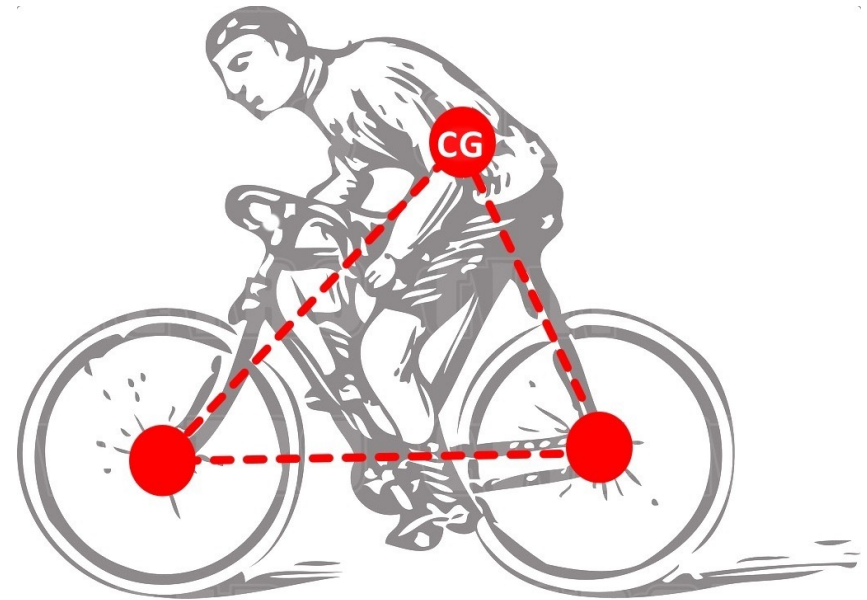
Gait change to gain more thrust

Walking Vs Running Vs Cycling

Running/Walking



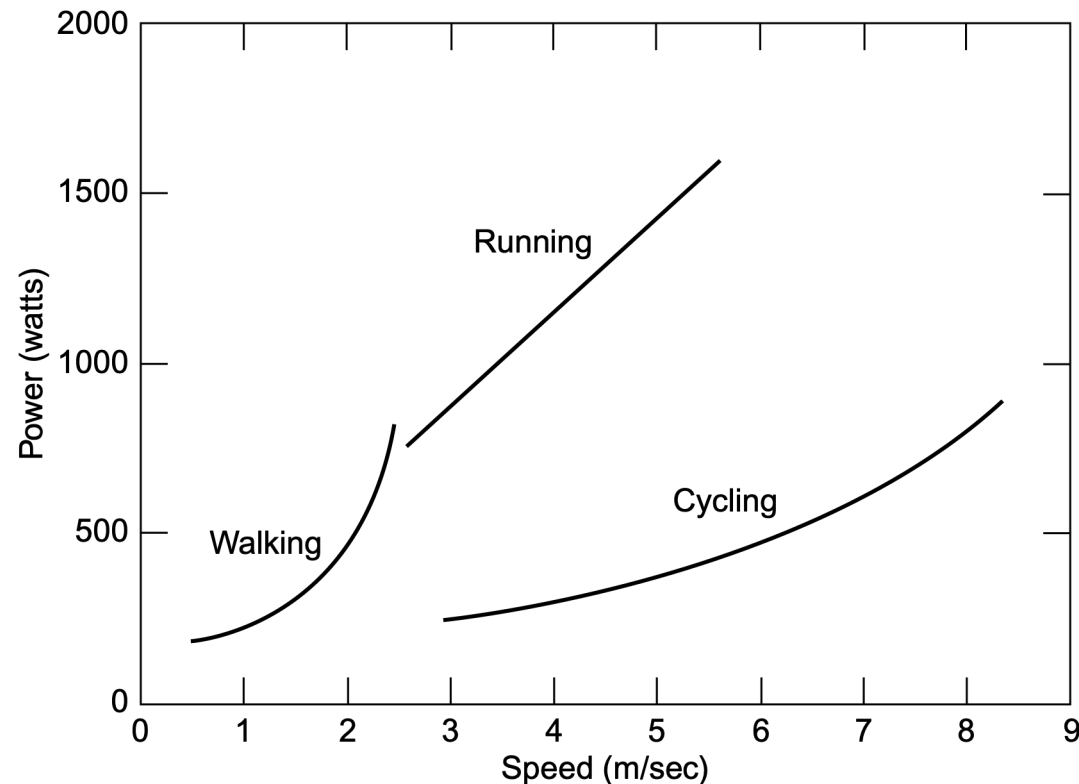
Cycling



<https://www.wheelfanatyk.com/blog/wheel-trick-three-braking/>

Variability in center of mass

Energy consumption



Running requires more power than cycling at the same speed, because, during cycling the center of gravity of the body remains almost at the same height.

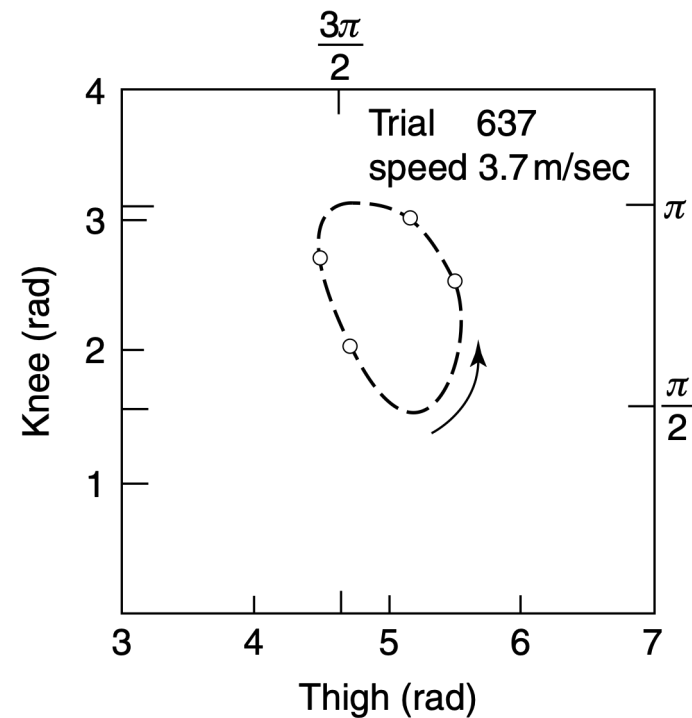
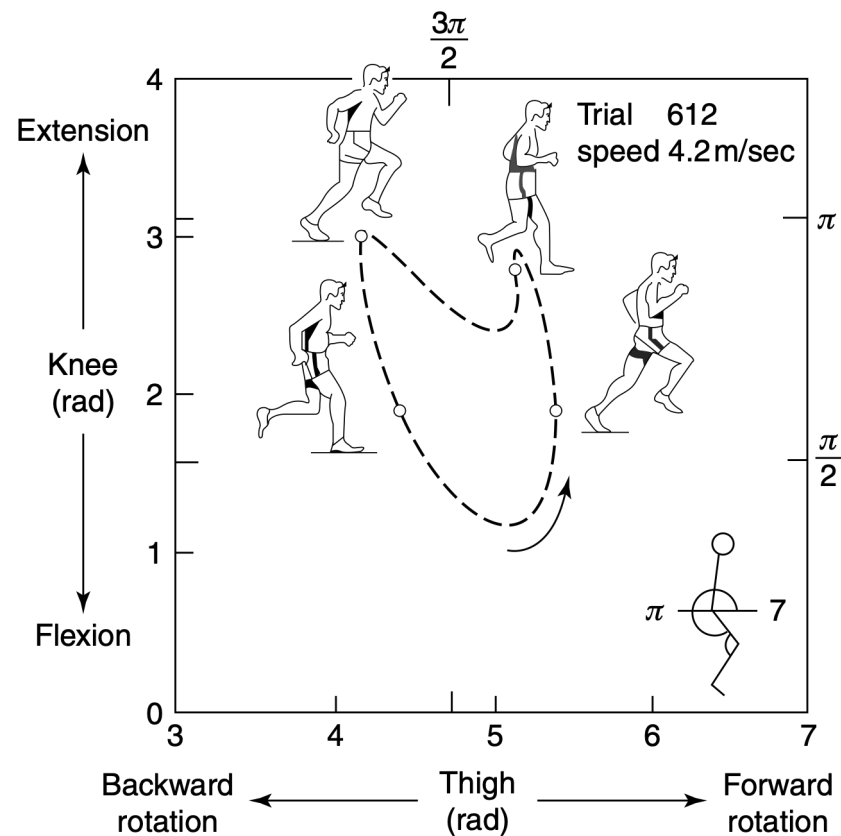
Point Walker stimuli

Gunnar Johansson created five-point walker stimuli by placing lights at joints of the subjects and filming the patterns of lights when subjects worked or carried out other actions in dark.



- Involved in perception of moods, intentions etc.
- Designing therapies for gait disorders
- Training with prosthetics
- Diagnosis of neurological disorders.

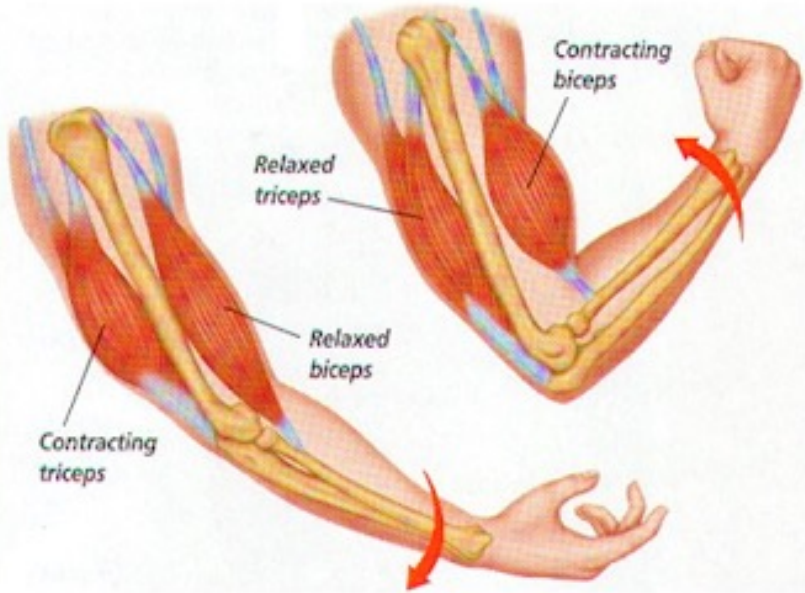
Angle-angle diagram



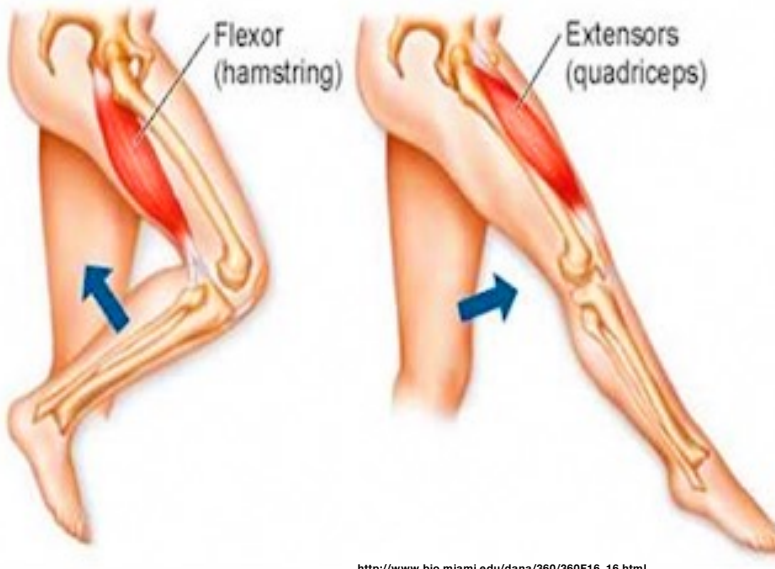
Neural control of locomotion

Role of nervous system?

Principle of reciprocity



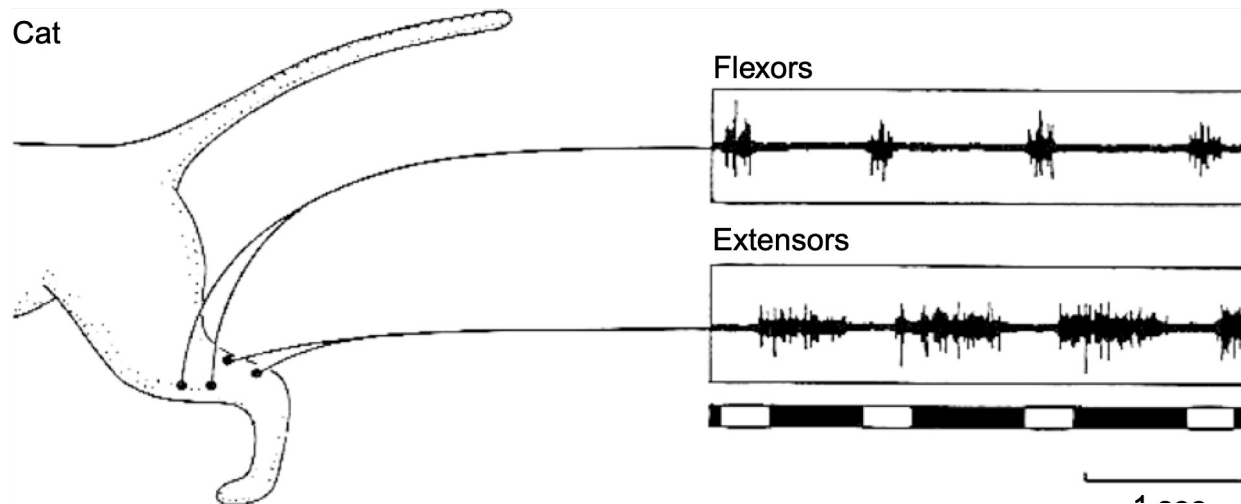
- Muscles tend to arrange in antagonist pairs
- To avoid counteracting each other function that could suppress movement



Sherrington law of reciprocal innervation

Muscular activity during walking

EMG Recording

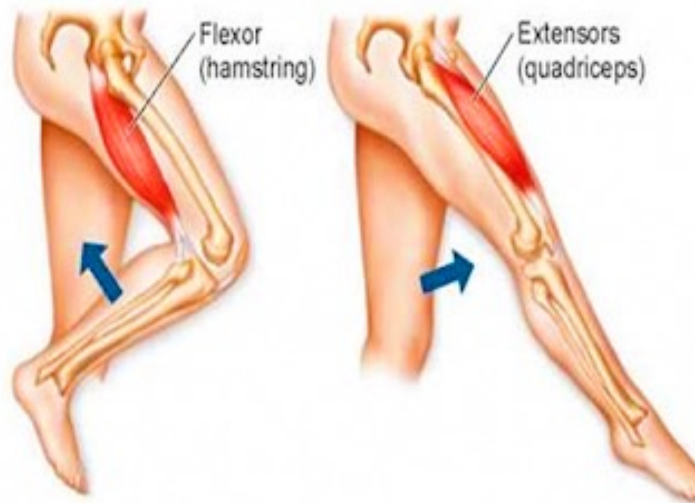


Swing phase

Flexor Active

Stance phase

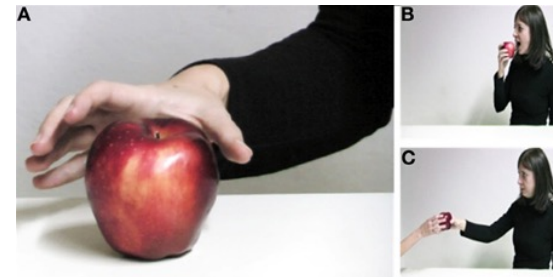
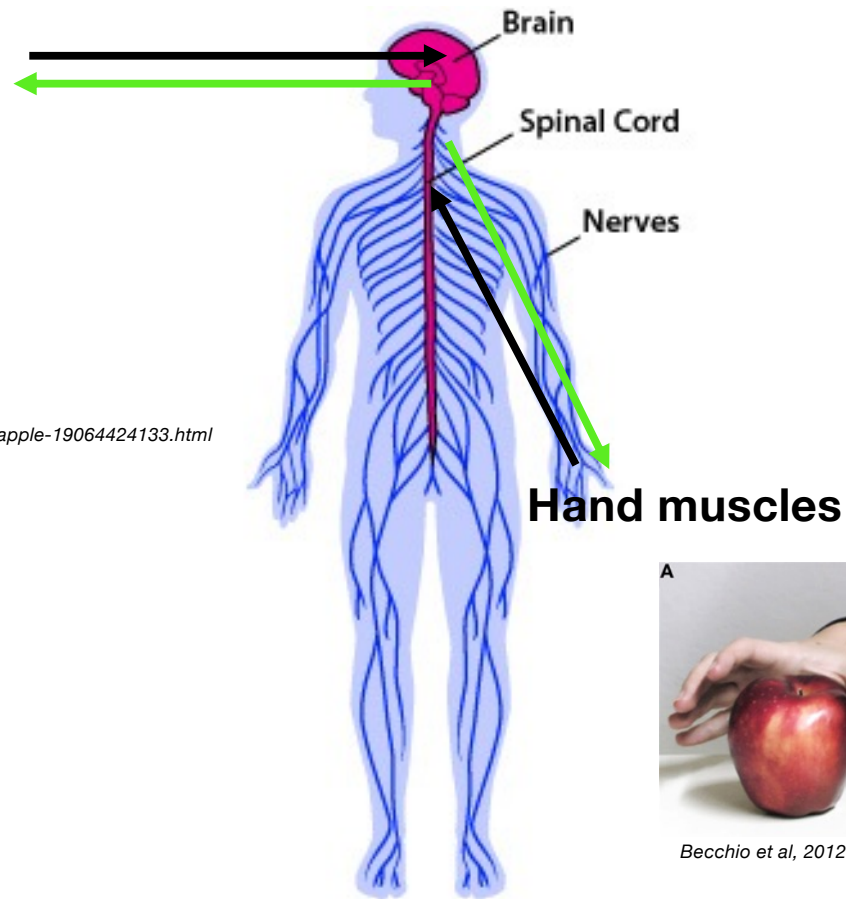
Extensors Active



The motor pathway



<https://www.indiamart.com/proddetail/fresh-apple-19064424133.html>



Becchio et al, 2012

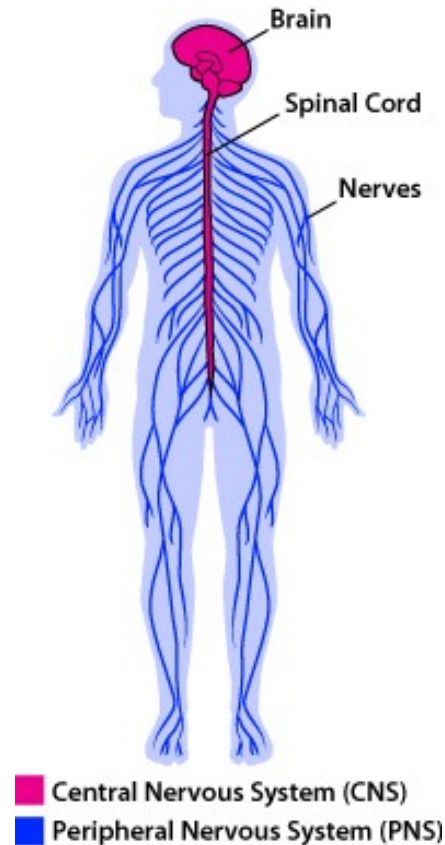
■ Central Nervous System (CNS)
■ Peripheral Nervous System (PNS)

Brain & Spinal Cord

Charles Scott Sherrington, 1906

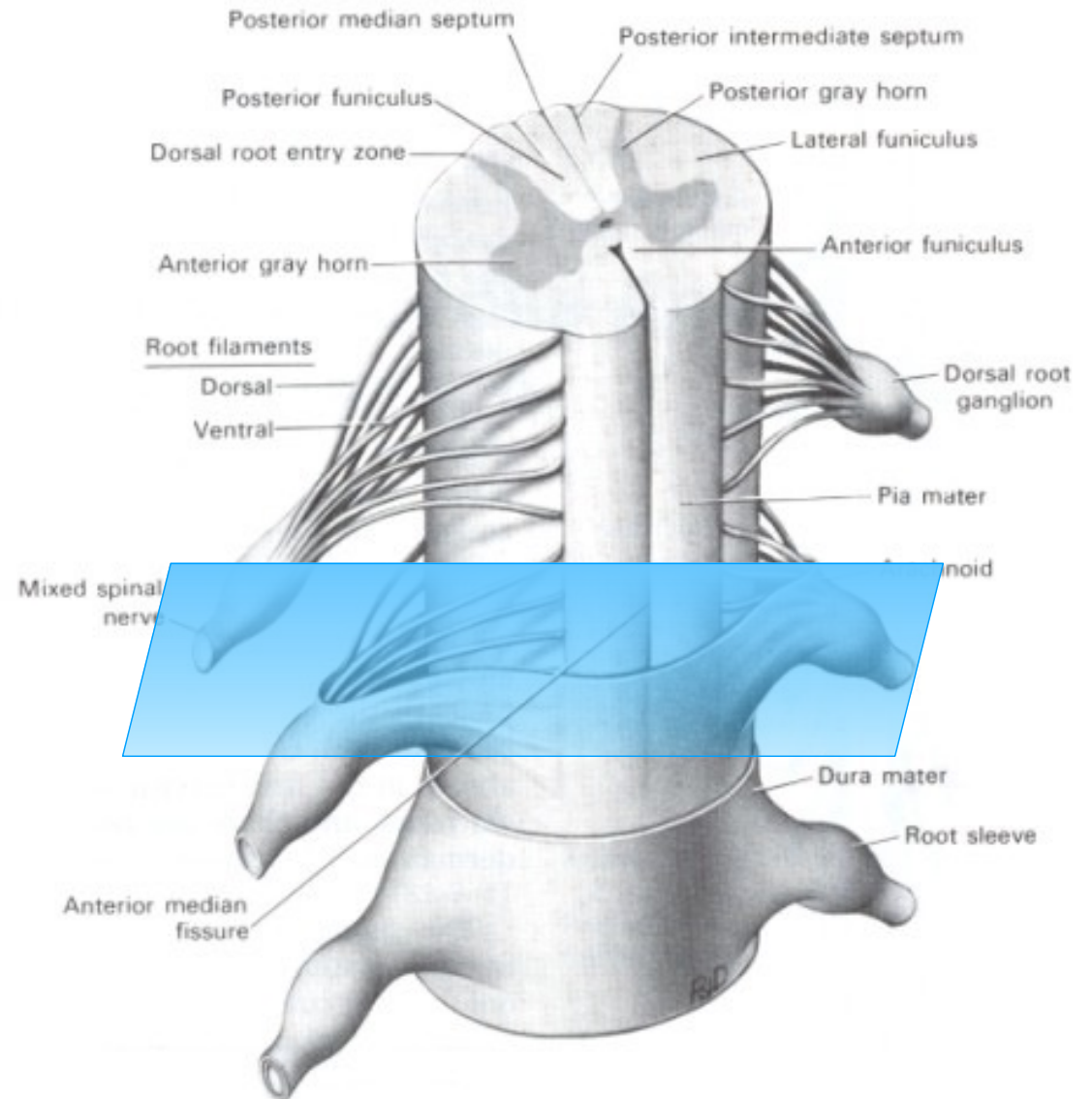
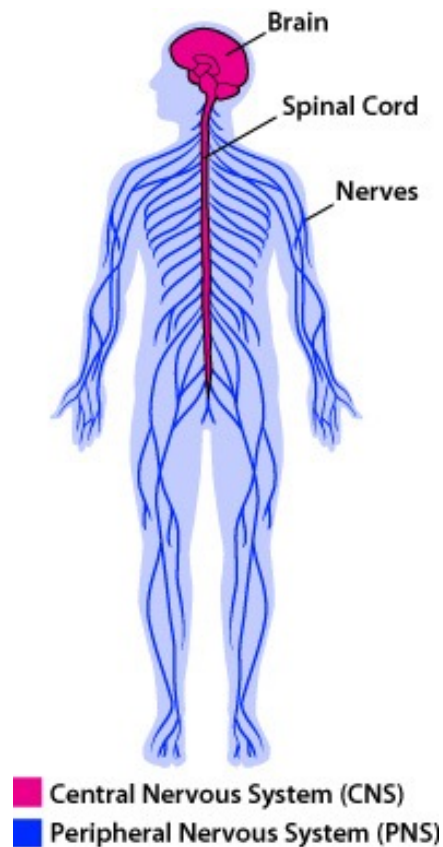


https://en.wikipedia.org/wiki/Charles_Scott_Sherrington



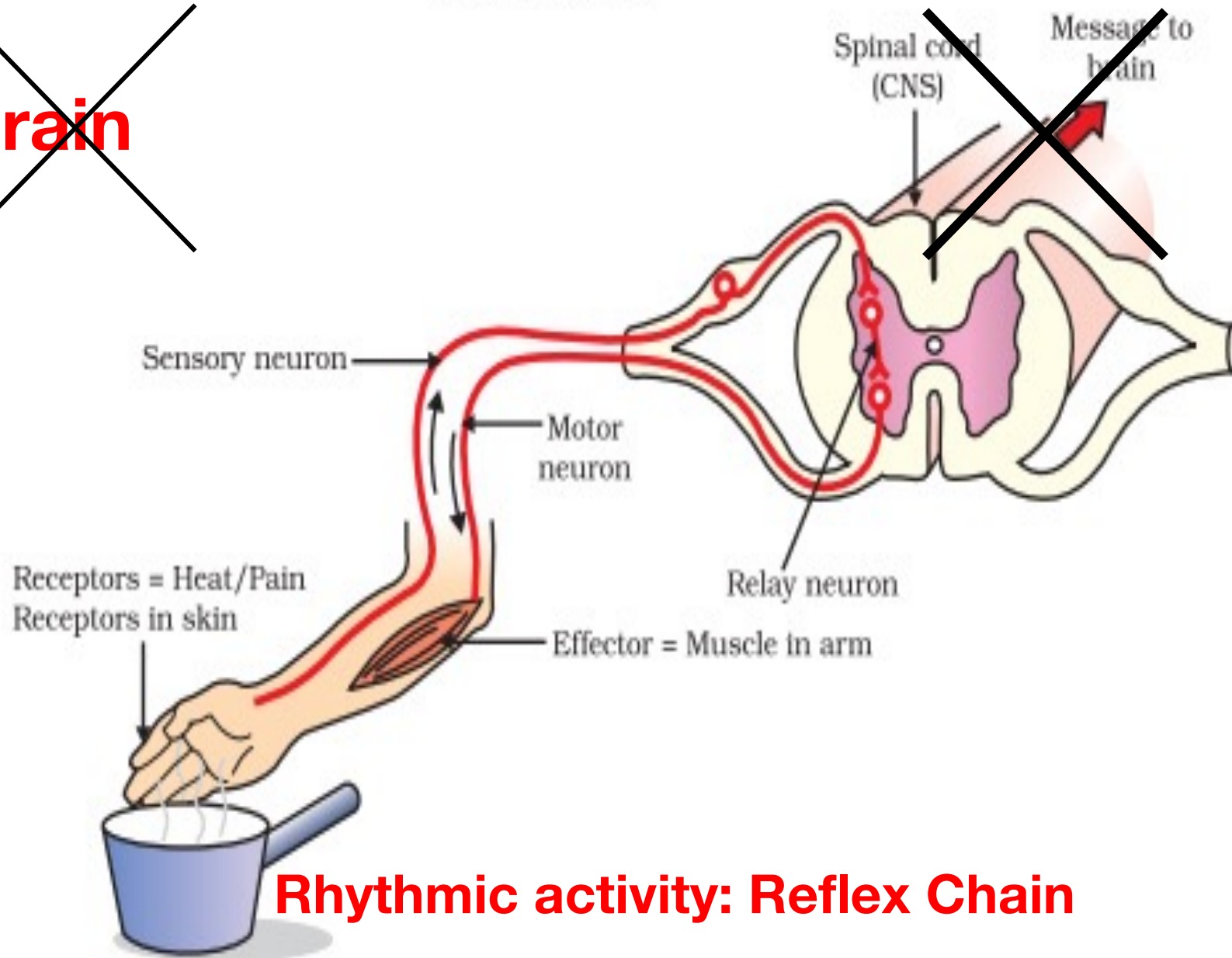
- Showed dissociation between brain and the spinal cord
- Spinal cord can produce rhythmic movement **without** input from the brain.

The Spinal Cord



Reflex Arc

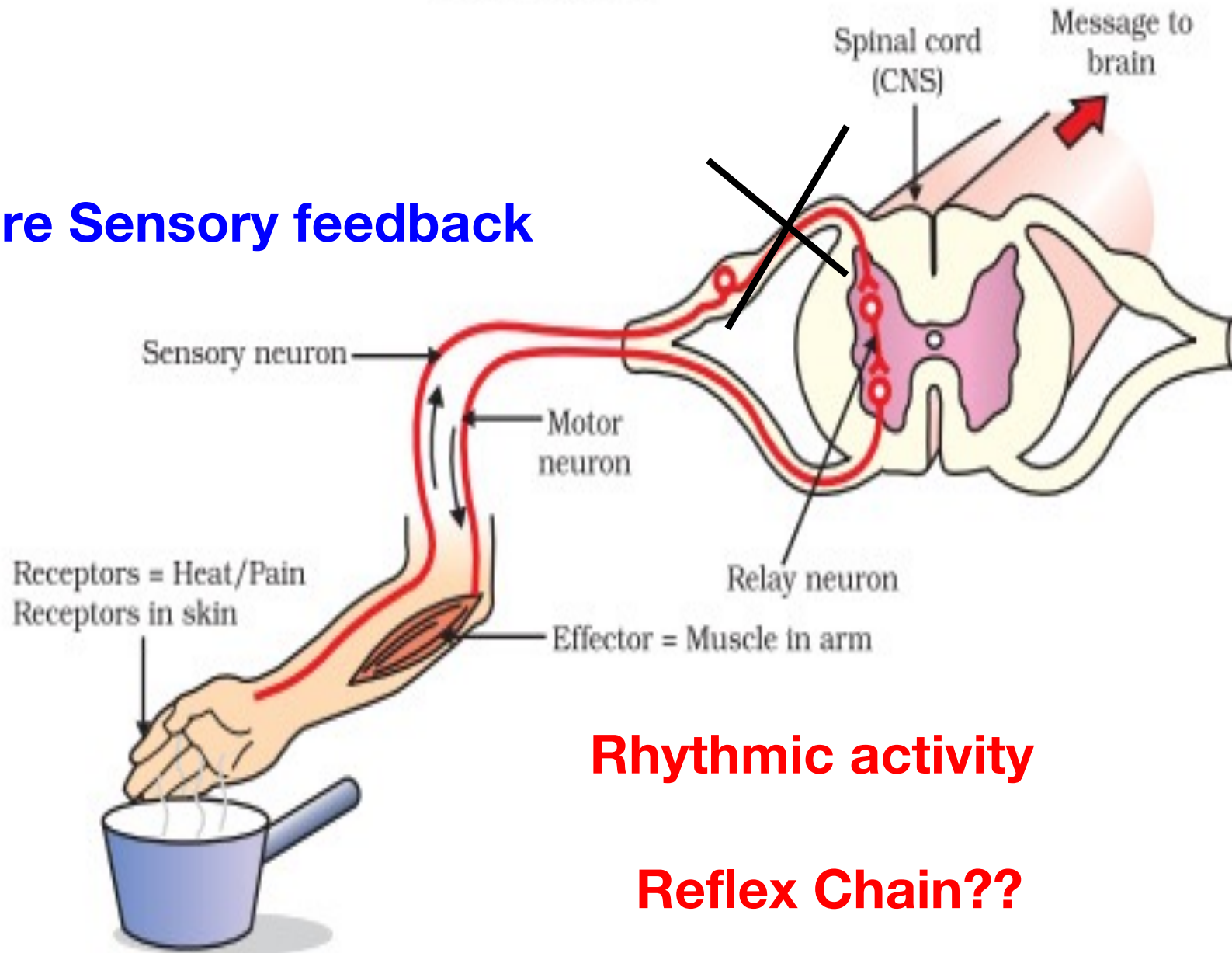
~~Brain~~



Rhythmic activity: Reflex Chain

Reflex Arc

Severe Sensory feedback



Severe Sensory feedback

Muscle paralyzed



Proprioception

Perception of information from receptors present in the muscles, joints, tendons and skin.

Kinesthesia

- Muscle spindles
- Golgi tendon organs
- Joint receptors
- Cutaneous receptors

